

The New Methods II

Matrix Mechanics: Forces, Moments, Torsion,
Stiffness, Least Squares, and the Z-Bead Extension

VSEPR-SIM thenewmethods

Everything below is undergraduate mechanics in different costumes, unified by a single observation: write the unknowns as a vector, build a system $\mathbf{Ax} = \mathbf{b}$, and solve.

Three-Force System

Unknown magnitudes L_6, L_{12}, L_{18} along known unit directions $\hat{u}_1, \hat{u}_2, \hat{u}_3$. Equilibrium against external load $\mathbf{P}$:

$$\underbrace{\begin{bmatrix} u_{1x} & u_{2x} & u_{3x} \\ u_{1y} & u_{2y} & u_{3y} \\ u_{1z} & u_{2z} & u_{3z} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} L_6 \\ L_{12} \\ L_{18} \end{bmatrix}}_{\mathbf{L}} = \underbrace{\begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix}}_{\mathbf{P}}$$

Ideal (exact geometry):

$$\mathbf{L} = \mathbf{A}^{-1}\mathbf{P}$$

Real (noisy / overdetermined):

$$\mathbf{L}_{\text{real}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P}$$

This covers cable tensions, strut forces, rod loads, ideal joint load sharing.

Add Moments About a Point

Each force $L_k \hat{u}_k$ has position vector $\mathbf{r}_k$ from point O . Stack force and moment equations:

$$\begin{bmatrix} \hat{u}_1 & \hat{u}_2 & \hat{u}_3 \\ \mathbf{r}_1 \times \hat{u}_1 & \mathbf{r}_2 \times \hat{u}_2 & \mathbf{r}_3 \times \hat{u}_3 \end{bmatrix} \begin{bmatrix} L_6 \\ L_{12} \\ L_{18} \end{bmatrix} = \begin{bmatrix} \mathbf{P} \\ \mathbf{M}_{\text{req}} \end{bmatrix}$$

(Each row of the upper block is a 3-vector; the full matrix is 6×3 .) Solve with the pseudo-inverse for a wrench balance.

Coordinate-Aligned Ideal vs. Real Example

Assume members aligned with coordinate axes: $\hat{u}_1 = (1, 0, 0)^T$, $\hat{u}_2 = (0, 1, 0)^T$, $\hat{u}_3 = (0, 0, 1)^T$, and load $\mathbf{P} = (120, 80, -40)^T$ N. Then $\mathbf{A} = \mathbf{I}$, so

$$\mathbf{L}_{\text{ideal}} = (120, 80, -40)^T \text{ N}$$

Negative sign: member 18 is in compression.

Real case – directions slightly off:

$$\mathbf{A} = \begin{bmatrix} 0.98 & 0.05 & 0.00 \\ 0.04 & 0.99 & 0.10 \\ 0.00 & 0.03 & 0.95 \end{bmatrix}$$

Solve $\mathbf{A}\mathbf{L} = \mathbf{P}$ for adjusted internal loads. Geometry error changes load sharing: what reality loves to do precisely when your design margin is already garbage.

Stiffness Method

Connect force and displacement directly:

$$\mathbf{K}\mathbf{u} = \mathbf{f}$$

$\mathbf{K}$ = stiffness matrix, $\mathbf{u}$ = nodal displacement vector, $\mathbf{f}$ = nodal force vector.

For a 1D two-spring system:

$$\begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}$$

Gateway to FEA; yields beam deflection, truss extension, support reactions.

Least-Squares / Pseudo-Inverse

Overdetermined or noisy system:

$$\mathbf{A}\mathbf{x} \approx \mathbf{b} \quad \implies \quad \mathbf{x} = \mathbf{A}^+\mathbf{b}$$

Residual and residual norm:

$$\mathbf{r} = \mathbf{b} - \mathbf{A}\mathbf{x} \quad \|\mathbf{r}\|$$

Useful for force reconstruction from sensor data, best-fit reaction estimates, and real-vs-ideal comparison.

Torsional System

Torsion identity:

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L}$$

Torsional stiffness of segment i :

$$k_i^{(t)} = \frac{G_i J_i}{L_i}$$

Element stiffness matrix for a two-node torsional element:

$$k_i^{(t)} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Assembled shaft system:

$$\mathbf{K}_\theta \boldsymbol{\theta} = \mathbf{T}$$

Post-solve, shear stress and angle of twist in each segment:

$$\tau_{\max,i} = \frac{T_i c_i}{J_i} \quad \phi_i = \frac{T_i L_i}{G_i J_i}$$

Unified Block System

Define the combined unknown vector:

$$\mathbf{q} = \begin{bmatrix} L_6 \\ L_{12} \\ L_{18} \\ \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

One block equation:

$$\begin{bmatrix} \mathbf{A}_f & \mathbf{0} \\ \mathbf{A}_m & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_\theta \end{bmatrix} \mathbf{q} = \begin{bmatrix} \mathbf{P} \\ \mathbf{M} \\ \mathbf{T} \end{bmatrix}$$

$\mathbf{A}_f$ = force balance, $\mathbf{A}_m$ = moment balance, $\mathbf{K}_\theta$ = torsion. One framework for force members, moment equilibrium, and rotational response.

Compact $L_{6,12,18}$ Notation Summary

$$\mathbf{L} = \begin{bmatrix} L_6 \\ L_{12} \\ L_{18} \end{bmatrix} \quad \mathbf{A}\mathbf{L} = \mathbf{P}$$

$$\mathbf{L}_{\text{ideal}} = \mathbf{A}^{-1}\mathbf{P} \quad \mathbf{L}_{\text{real}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P} \quad \mathbf{r} = \mathbf{P} - \mathbf{A}\mathbf{L}_{\text{real}}$$

Exact solution, best-fit solution, error measure. Neat.

Z -Bead Extension: Forces from First Principles

Each particle i carries state:

$$Z_i = \begin{bmatrix} \mathbf{x}_i \\ \mathbf{v}_i \\ s_i \\ q_i \\ m_i \\ \varepsilon_i \end{bmatrix} \quad \mathbf{Z} = \{Z_1, Z_2, \dots, Z_N\}$$

Force system becomes Z -dependent:

$$\mathbf{A}(\mathbf{Z}) \mathbf{L} = \mathbf{P}(\mathbf{Z})$$

Pairwise interaction:

$$\mathbf{F}_{ij} = f(s_i, s_j, r_{ij}, q_i, q_j, \nabla \Phi) \quad r_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$$

Total force on bead i :

$$\mathbf{F}_i = \sum_{j \neq i} \mathbf{F}_{ij}$$

Macro Property Extraction

$$\rho = \frac{\sum_i m_i}{V(\mathbf{Z})} \quad \boldsymbol{\sigma} = \frac{1}{V} \sum_i \mathbf{F}_i \otimes \mathbf{x}_i \quad E = \sum_i \varepsilon_i + \sum_{i < j} U_{ij}$$

Ideal vs. Real in Z -Language

$$\text{Ideal: } \mathbf{Z}_{\text{ideal}} \Rightarrow \mathbf{A}(\mathbf{Z})^{-1} \mathbf{P}(\mathbf{Z})$$

$$\text{Real: } \mathbf{L}_{\text{real}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{P}(\mathbf{Z}_{\text{real}}) \quad \mathbf{r} = \mathbf{P} - \mathbf{A}(\mathbf{Z}) \mathbf{L}$$

Coarse-Grained Force Channels

Each channel $L_k(\mathbf{Z})$ is a projection of many bead forces:

$$L_k(\mathbf{Z}) = \sum_{i \in \Omega_k} \mathbf{F}_i \cdot \hat{u}_k$$

Unified System

$$\begin{cases} \mathbf{F}(\mathbf{Z}) = \mathbf{A}(\mathbf{Z})\mathbf{L} \\ \mathbf{T}(\mathbf{Z}) = \mathbf{K}_\theta(\mathbf{Z}) \boldsymbol{\theta} \\ \mathbf{P}(\mathbf{Z}) \rightarrow \text{macro properties} \end{cases}$$

Advanced Methods from the Same Framework

Goal	Method
Natural modes / stability	$\mathbf{K}\mathbf{v} = \lambda\mathbf{v}$ (eigenvalue problem)
Best-fit load estimate	$\min_{\mathbf{x}} \ \mathbf{A}\mathbf{x} - \mathbf{b}\ ^2$
Design under equilibrium	$\min J(\mathbf{x})$ subject to $\mathbf{A}\mathbf{x} = \mathbf{b}$
Numerical stability check	$\kappa(\mathbf{A})$ (condition number)
Sensor fusion	$\mathbf{L} = \mathbf{A}^+ \mathbf{P}_{\text{sensor}}$

From force balance through eigenvalue stability through bead dynamics, it is the same algebraic skeleton wearing different physics.